

# WHAT MAKES A SONG TRAVEL?

## Audio Features vs. Language Barriers in Global Music Diffusion

**Course:** DSA210 Spring'26 Individual Project

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**Date:** May 17, 2026

**Dataset Scale:** 299,958 Observations (598 Artists)

### 1. MOTIVATION

In the modern streaming era, music has become increasingly globalized. However, the mechanism driving a song's cross-border transition remains a critical point of debate: does a song's international success depend primarily on its cultural and linguistic accessibility, or can an exceptional technical audio structure enable it to bypass traditional linguistic barriers?

This research has two complementary objectives. First, it investigates whether high-performing audio characteristics (e.g., energy, tempo, danceability) can generate an "Audio Signature" strong enough to overcome severe language barriers. Second, it isolates and ranks individual musical features to identify the primary technical drivers of cross-border speed once linguistic advantages are statistically controlled. By analyzing "Migration Speed," this study maps out how structural sonic elements and structural language gaps interact to dictate global chart trajectories.

### 2. DATA SOURCE & COLLECTION

The research integrates corporate streaming metadata with standard dyadic country-level linguistic metrics to build a comprehensive analytical database. The final merged dataset contains **299,958 historical records across 598 distinct artists**. The foundational sources include:

- **Historical Spotify Charts (charts.csv):** Gathered via Kaggle (compiled originally by Dhruvil Dave), providing daily and weekly regional chart rankings across more than 35 countries.
- **Spotify Audio Features (Spotify\_Dataset\_V3.csv):** Gathered via Kaggle (compiled by Bruno Alarcon), containing tracks' granular audio metadata extracted via the Spotify API. This includes numeric track elements (Danceability, Energy, Loudness, Speechiness, Valence, Acousticness, and Instrumentalness) normalized from 0 to 1.
- **Melitz & Toubal (2014) Linguistic Dataset (ling\_web.dta):** A recognized geopolitical database from CEPII that maps country pairs (dyads) to deliver structural linguistic metrics: Common Spoken Language (CSL) and Linguistic Proximity (LP).

**Data Preprocessing & Processing Pipeline:** Data pipeline construction was handled in Python using the Pandas library. Streaming data rows were mapped to linguistic indicators through ISO 3-digit country codes. For each song, the framework established a baseline market debut date to capture cross-border velocity, calculating the **Migration Lag (days)** as the temporal delta between a song's domestic debut and its subsequent appearance on a foreign market's Top 200 chart.

CRITICAL DATA CHARACTERISTIC: 71.7% of the tracking observations are right-censored. This indicates that the vast majority of tracks do not complete their full international diffusion cycle during the observation period—they remain active or fail to enter specific target markets entirely. Traditional Ordinary Least Squares (OLS) regressions completely fail under these conditions, introducing severe survival bias into the parameters.

To handle this structural issue, a specialized **Barrier Score** was created as a planned enrichment to measure total cross-border impedance:

$$\text{Barrier Score} = 1 - f(\text{CSL}, \text{LP})$$

This synthetic scale models the mutual comprehensibility gap and language family distance, capturing prominent global English market advantages alongside complex localized corridors (e.g., Turkish music within Germany or Spanish tracks within Italy).

### 3. DATA ANALYSIS METHODOLOGY

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The project used a multi-layered econometric and machine learning framework designed to circumvent right-censoring bias and control for structural groupings:

1. **Survival Analysis (Cox Proportional Hazards Model):** Using the `lifelines` library, a regularized Cox PH model ( $\lambda = 0.1$ ) was deployed to model Migration Lag without discarding right-censored data. The model computes the hazard ratio ( $\exp(\text{coef})$ ) to evaluate how specific features scale or restrict diffusion speed.
2. **Linear Mixed-Effects Modeling (MixedLM):** Built via `statsmodels` using  $\log(\text{Migration Lag})$  as the target variable. This approach incorporates **Artist-Level Random Effects** to isolate individual star power and promotional budgets, ensuring that structural baseline differences across artists do not distort the fixed-effect audio coefficients.
3. **Advanced Machine Learning Pipeline:** An **XGBoost Classifier** was built to predict "Fast Hits" (defined as an international market transition occurring in under 14 days). Model interpretability was established through **SHAP (SHapley Additive exPlanations)** global feature importance and directional dependency profiling.
4. **Geopolitical Network Analysis:** Built via `networkx` using cross-border music migrations to structure a directional network graph. Centrality indexes (PageRank and Betweenness Centrality) were extracted to locate dominant receiving markets and crucial international relay bridges.

## 4. EMPIRICAL FINDINGS

### A. Survival and Mixed-Effects Model Estimations

The table below provides a comprehensive overview of the econometric models. A significance audit reveals that traditional OLS estimates suffer from false positives due to survival censoring, notably misrepresenting the impact of Loudness.

| COVARIATE<br>FEATURE | MODEL 1: REGULARIZED COX<br>PROPORTIONAL HAZARDS |                 |             | MODEL 2: MIXED LINEAR MODEL<br>(ARTIST RANDOM EFFECTS) |         |         | ROBUSTNESS<br>VERDICT    |
|----------------------|--------------------------------------------------|-----------------|-------------|--------------------------------------------------------|---------|---------|--------------------------|
|                      | COEF                                             | HAZARD<br>RATIO | P-<br>VALUE | COEF                                                   | Z-SCORE | P-VALUE |                          |
| Barrier Score        | -0.591                                           | 0.554           | 0.000       | +1.209                                                 | 28.715  | 0.000   | ✓ Robust<br>Barrier      |
| Danceability         | -0.304                                           | 0.738           | 0.000       | -1.428                                                 | -40.664 | 0.000   | ✓ Robust<br>Speed Driver |
| Energy               | +0.174                                           | 1.190           | 0.000       | +0.896                                                 | 12.597  | 0.000   | ✓ Robust<br>Inhibitor    |
| Loudness             | -1.794                                           | 0.166           | 0.000       | -0.056                                                 | -1.704  | 0.088   | ⚠ OLS False<br>Positive  |
| Speechiness          | +0.708                                           | 2.029           | 0.000       | +1.340                                                 | 43.838  | 0.000   | ✓ Robust Delay<br>Factor |
| Valence              | +0.126                                           | 1.134           | 0.000       | +0.869                                                 | 42.916  | 0.000   | ✓ Robust Delay<br>Factor |
| Acousticness         | +0.616                                           | 1.852           | 0.000       | +0.478                                                 | 25.226  | 0.000   | ✓ Robust Delay<br>Factor |
| Instrumentalness     | -0.604                                           | 0.547           | 0.000       | +1.397                                                 | 23.232  | 0.000   | ✓ Robust<br>Divergence   |
| Energy × Barrier     | +0.002                                           | 1.002           | 0.924       | -1.132                                                 | -16.115 | 0.000   | ✓ Robust<br>Interaction  |

## CORE ECONOMETRIC INSIGHTS

- **Linguistic Barriers Dominate:** The Cox Hazard Ratio for the Barrier Score is **0.554**. This reveals that a maximum language barrier reduces a track's international migration probability by **44.6%** at any given point in time compared to a zero-barrier context. In the MixedLM framework, higher barrier scores show a large, highly significant positive coefficient (**+1.209,  $z = 28.715$** ), confirming that structural linguistic differences act as a substantial drag on migration speed.
- **The Audio Signature:** High **Danceability** significantly shortens travel times, exhibiting a strong negative impact on migration lag (**-1.428,  $z = -40.664$** ). Conversely, extreme **\*\*Energy\*\*** and high **\*\*Speechiness\*\*** (**+1.340**) systematically slow down diffusion. This indicates that lyric-heavy and highly chaotic compositions struggle to travel cross-border, whereas rhythmically stable, highly danceable tracks function as universal vectors.
- **The Interaction Effect:** The **Energy × Barrier Score** interaction is highly significant in the MixedLM framework (**-1.132,  $z = -16.115$** ). This reveals a critical structural phenomenon: while high energy generally slows down global migration in open, low-barrier markets, it changes direction when confronting strict language barriers. High-energy productions actively mitigate a portion of the linguistic drag, serving as a mechanical compensation vector.

## B. Machine Learning Predictive Power & Interpretability

The predictive framework demonstrates high classification accuracy and clear regional patterns:

- **Model Benchmarking:** While a simple baseline yields an accuracy of only **0.517**, the Random Forest model achieves an accuracy of **0.747**. The optimized **XGBoost Classifier achieves an exceptional ROC-AUC of 0.938 and a PR-AUC of 0.938**. Cross-validation verifies excellent model stability ( **$0.933 \pm 0.003$** ), with an optimal classification threshold of **0.529** yielding a peak F1-score of **0.854**. In continuous regression setups, the framework achieves an  **$R^2$  of 0.641** with a Mean Absolute Error (MAE) of  **$\pm 19.1$  days**.
- **SHAP Summary Insights:** SHAP global feature and dot plots confirm that the Barrier Score exerts the strongest overall influence on migration speed. However, they also reveal a pronounced non-linear relationship: once an audio asset exceeds an explicit threshold of rhythm and danceability, its SHAP value shifts rapidly in a negative direction, systematically overriding mild linguistic friction.

## C. Macro Case Studies & Counterfactual Simulations

The model was subjected to counterfactual scenarios to isolate explicit industry trends:

- **The K-Pop Paradox:** Globally, the average migration lag across the dataset is **36.0 days**, with an average baseline Barrier Score of **0.907**. For K-Pop, the observed average migration lag rises sharply to **64.5 days** due to maximum linguistic distance (Barrier Score = **1.000**). In a counterfactual simulation where the language barrier is set to zero, the model predicts an unexpected rise in lag to 30.8 days relative to its rapid sub-regional baseline of 11.2 days. Diagnostic validation confirms that while K-Pop moves exceptionally fast within specialized regional fan networks, maximum linguistic barriers slow down its

broader global expansion; the powerful fandom effect is real but remains constrained outside its primary geographic corridors.

- **The Latin Highway & Nordic Sync Corridors:** The simulations confirm that regional and cultural proximity act as strong historical cushions. Shared cultural and linguistic roots save an average of **3.1 days** along the Latin Highway and **5.2 days** within the interconnected Nordic Sync network.

D. Network Analysis: Music Relay Hubs

The structural network analysis isolates clear geographical imbalances between dominant consumer targets and strategic transit bridges:

Top 5 Music Receiving Hubs (PageRank)

| ISO | COUNTRY NAME | PAGERANK | IN-DEGREE | OUT-DEGREE |
|-----|--------------|----------|-----------|------------|
| CHE | Switzerland  | 0.0418   | 36        | 0          |
| AUT | Austria      | 0.0254   | 34        | 29         |
| SWE | Sweden       | 0.0237   | 30        | 67         |
| SVK | Slovakia     | 0.0236   | 36        | 0          |
| DEU | Germany      | 0.0213   | 30        | 65         |

Top 5 Transit Relay Countries (Betweenness)

| ISO | COUNTRY NAME | BETWEENNESS | TOTAL CONNECTIONS | AVG SPEED  |
|-----|--------------|-------------|-------------------|------------|
| MAR | Morocco      | 0.1582      | 88                | 247.6 days |
| ITA | Italy        | 0.0731      | 98                | 79.1 days  |
| AUT | Austria      | 0.0707      | 63                | 14.7 days  |
| PHL | Philippines  | 0.0586      | 85                | 3.2 days   |
| LTU | Lithuania    | 0.0575      | 65                | 93.3 days  |

**Network Structure Interpretation:** The network analysis reveals a clear distinction between consumer endpoints and global transmission gears. **Switzerland (CHE)** represents the ultimate global receiving sink; due to its multi-lingual structure (French, German, Italian), it absorbs content rapidly from multiple distinct language families, resulting in a high PageRank (**0.0418**), a high in-degree (**36**), and zero outward transmission. Conversely, **Morocco (MAR)** exhibits the highest global Betweenness Centrality (**0.1582**) over 88 major connections, despite a long average transmission delay of 247.6 days. Morocco acts as a strategic transatlantic cultural bridge, linking Arabic, French, and Spanish repertoires to relay music across distinct geopolitical clusters.

## 5. LIMITATIONS & FUTURE WORK

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### ANALYTICAL CONSTRAINTS

While the empirical framework handles survival censoring and artist random effects, several structural gaps remain:

- **Platform and Consumption Bias:** The study relies exclusively on Spotify chart indexes, which heavily reflect urban demographics and younger listener groups. It lacks coverage of regional streaming choices (e.g., Tencent Music in China or YouTube Music in developing markets), which limits its generalizability.
- **Omission of External Micro-Drivers:** The data models cannot capture external marketing forces, such as viral TikTok algorithms, localized algorithmic playlist recommendations, or direct music label financing. These omitted parameters explain a significant portion of the remaining model variance.

### FUTURE EXTENSIONS

- **Dynamic Temporal Deep Learning:** Future iterations should replace static classification architectures with recurrent networks (RNNs) or long short-term memory (LSTM) models to capture shifting chart trajectories over time.
- **Integration of Global Marketing Data:** Incorporating cross-platform social media metrics and streaming service playlist allocations would help isolate structural marketing drivers from pure audio characteristics.